Desk controller

Table of contents

1 Desk controller
2 System_overview
3 Interface_Power_supply
4 Interface_servomotor
5 Interface_Limit_switches
6 Block_diagram
7 DCDC_converter_3V3
8 Microcontroller
9 Servomotor_interface
10 Limit_switch_interface

Change history

1.0.0	23/09/2026	First release
1.1.0	16/10/2026	Migration to FLRHW KiCad starter template

System overview

File: System_overview.kicad_sch

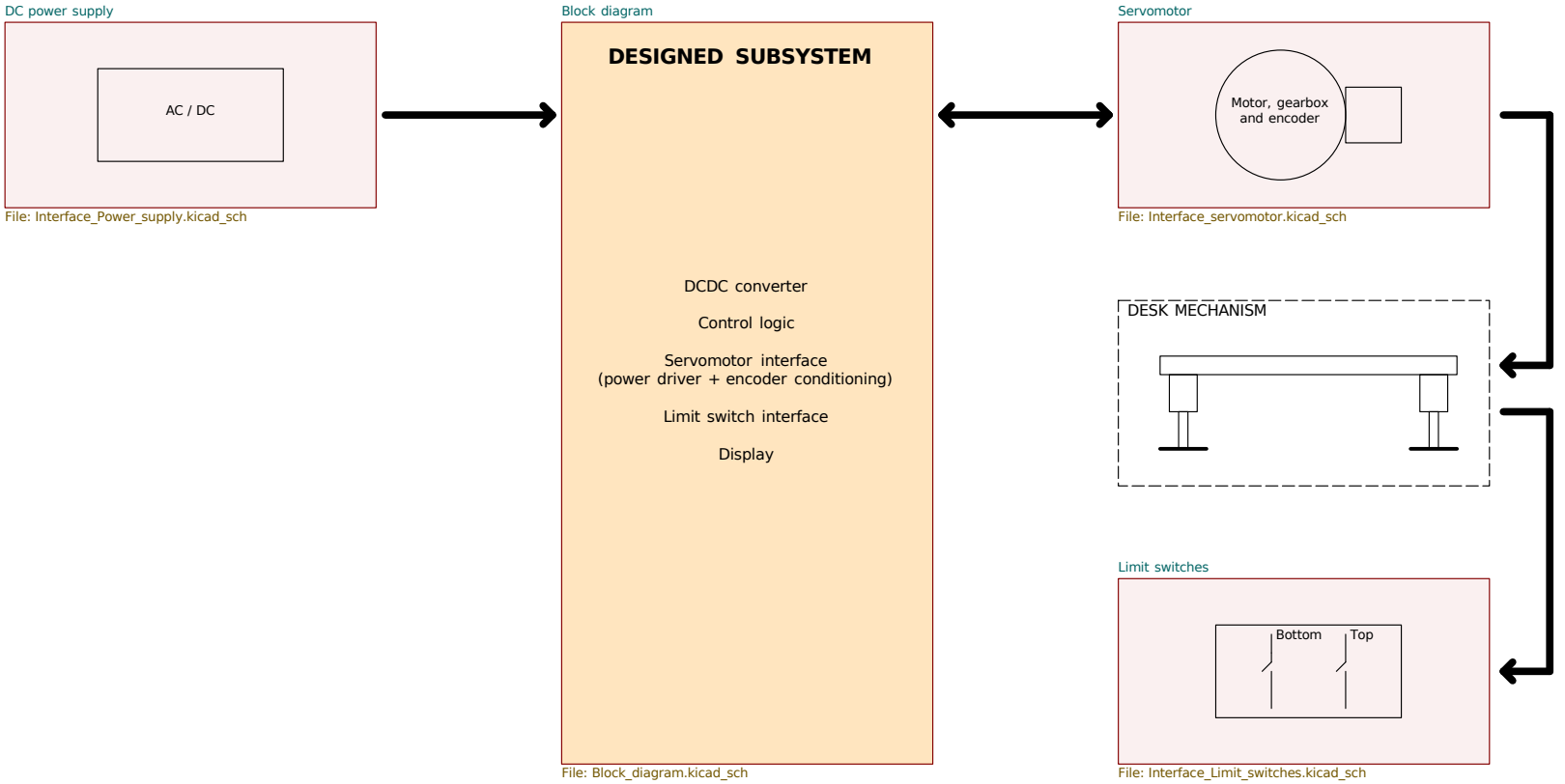
ChurrasCod Electronics

Desk controller

Notes

Sheet: Root	
File: Desk_controller.kicad_sch	
Rev: 1.1.0	Drwn: FLR
Mat: Concept	Rev'd: FLR
Date: 2026-10-18	Rls'd: FLR
Dwg: NA	Size: A4
Generated with KiCad	Page: 1/11

System overview



This sheet shows the overall system and the interfaces to external components.
External components are shown for system context only and are not part of this schematic.
Only the "DESIGNED SUBSYSTEM" is covered by this schematic.

Return to front sheet

ChurrasCod Electronics

Desk controller

Notes

Sheet: System overview	
File: System_overview.kicad_sch	
Rev: 1.1.0	Drwn: FLR
Mat: Concept	Rev'd: FLR
Date: 2026-10-18	Rls'd: FLR
Dwg: NA	Size: A4
Generated with KiCad	Page: 2/11

DC power supply

Interface specification

Barrel connector (Dell type, with ID pin)

Pin	
1	Supply - 19.5 V, 3 A (inner shell)
2	ID (internal pin)
3	GND (outer shell)
SH	Shield

Return to front sheet

ChurrasCod Electronics

Desk controller

Notes

Sheet: DC power supply	
File: Interface_Power_supply.kicad_sch	
Rev: 1.1.0	Drwn: FLR
Mat: Concept	Rev'd: FLR
Date: 2026-10-18	Rls'd: FLR
Dwg: NA	Size: A4
Generated with KiCad	Page: 3/11

Servomotor

Interface specification

JST B6B-XH-A / JST XHP-6 connector

Pin	Description
1	Encoder B output (VCC)
2	Encoder A output (VCC)
3	GND
4	VCC (3.3 - 5V)
5	Motor B (3 A, 12-24 V)
6	Motor A (3 A, 12-24 V)

Return to front sheet

ChurrasCod Electronics

Desk controller

Notes

Sheet: Servomotor	
File: Interface_servomotor.kicad_sch	
Rev: 1.1.0	Drwn: FLR
Mat: Concept	Rev'd: FLR
Date: 2026-10-18	Rls'd: FLR
Dwg: NA	Size: A4
Generated with KiCad	Page: 4/11

Limit switches

Interface specification

JST B4B XH A / JST XHP-4 connector

Pin	Description
1	Bottom limit switch (Normally open, switch closes to GND)
2	GND
3	GND
4	Top limit switch (Normally open, switch closes to GND)

Return to front sheet

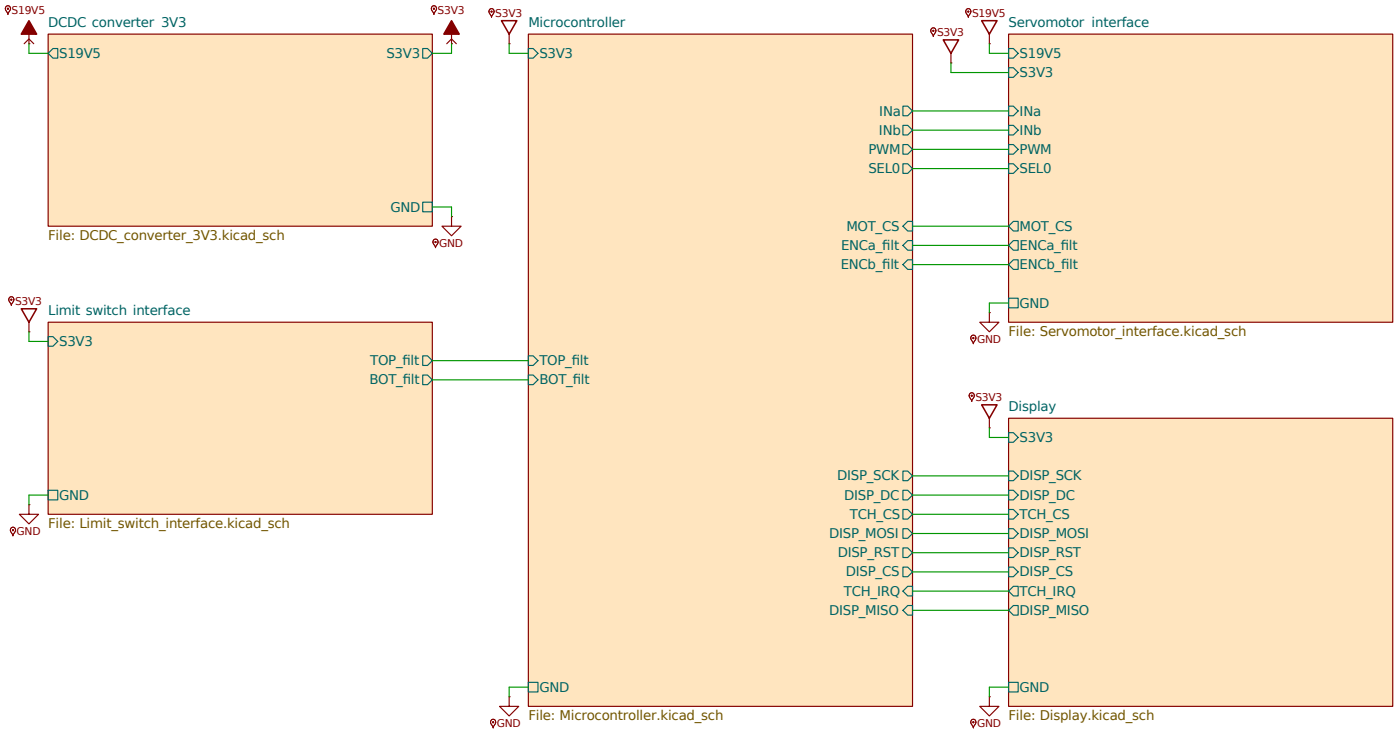
ChurrasCod Electronics

Desk controller

Notes

Sheet: Limit switches	
File: Interface_Limit_switches.kicad_sch	
Rev: 1.1.0	Drwn: FLR
Mat: Concept	Rev'd: FLR
Date: 2026-10-18	Rls'd: FLR
Dwg: NA	Size: A4
Generated with KiCad	Page: 5/11

Block diagram



Return to front sheet

ChurrasCod Electronics

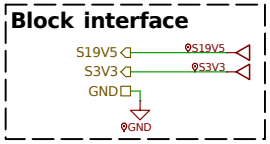
Desk controller

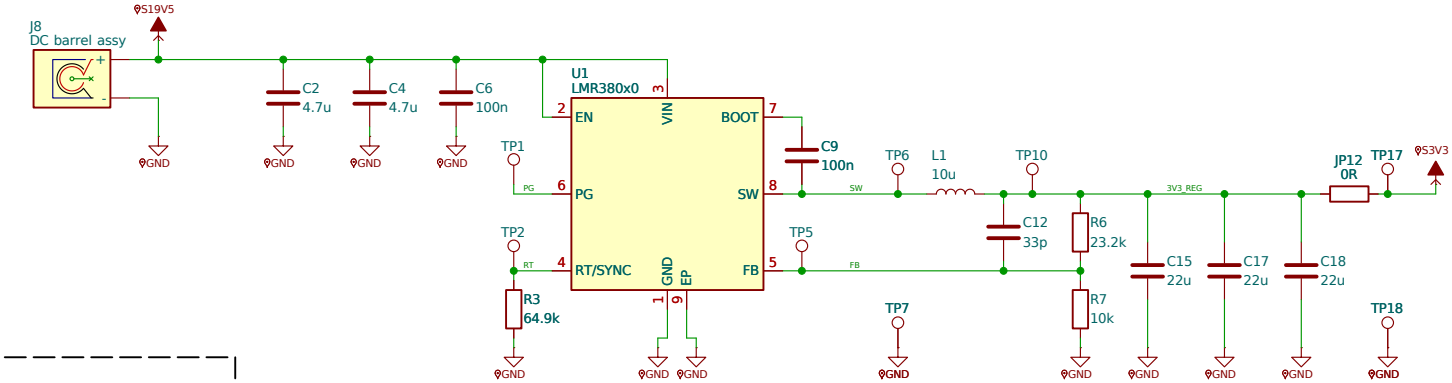
Notes

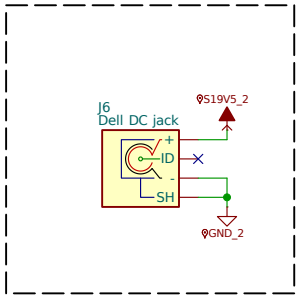
Sheet: Block diagram	
File: Block_diagram.kicad_sch	
Rev: 1.1.0	Drwn: FLR
Mat: Concept	Rev'd: FLR
Date: 2026-10-18	Rls'd: FLR
Dwg: NA	Size: A4
Generated with KiCad	Page: 6/11

DCDC converter 3V3

Block interface







J6 is fitted to a small "standalone" PCB which will then be soldered into the "main" PCB.
This standalone PCB + connector are what make the "DC barrel assy" used in J8 position.

Return to front sheet

ChurrasCod Electronics

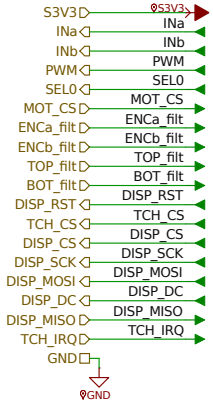
Desk controller

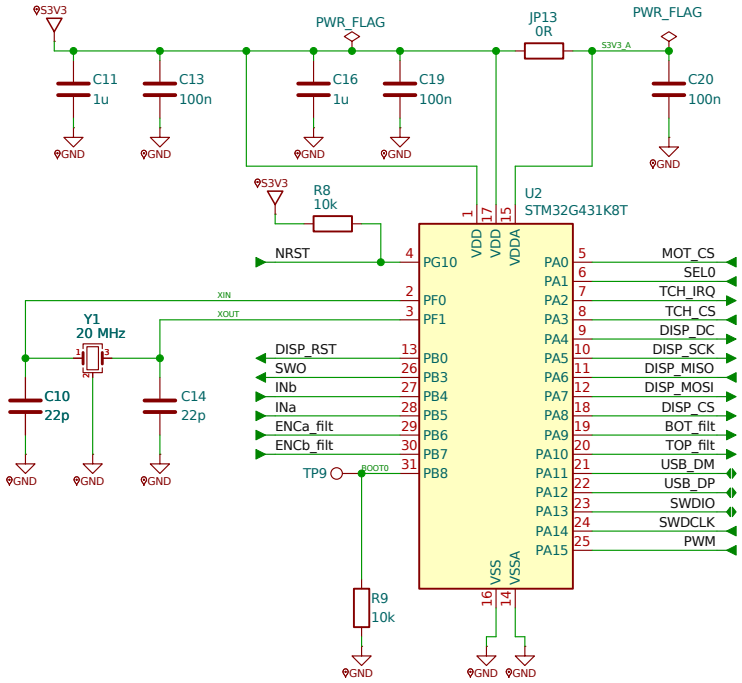
Notes

Sheet: DCDC converter 3V3	
File: DCDC_converter_3V3.kicad_sch	
Rev: 1.1.0	Drwn: FLR
Mat: Concept	Rev'd: FLR
Date: 2026-10-18	Rls'd: FLR
Dwg: NA	Size: A4
Generated with KiCad	Page: 7/11

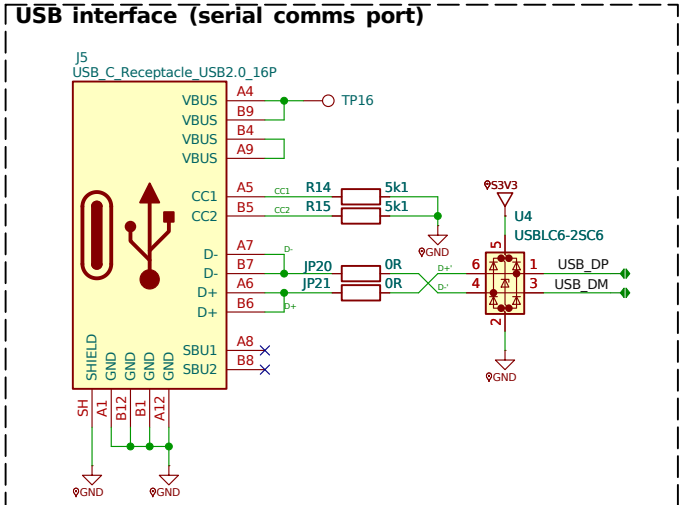
Microcontroller

Block interface

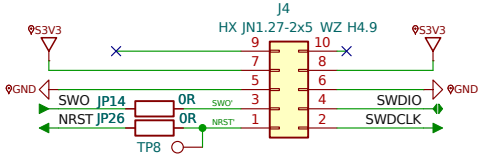




USB interface (serial comms port)



SWD header



Return to front sheet

ChurrasCod Electronics

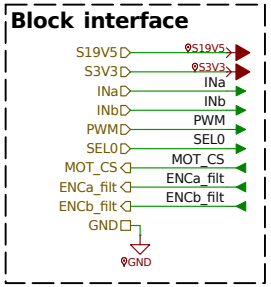
Desk controller

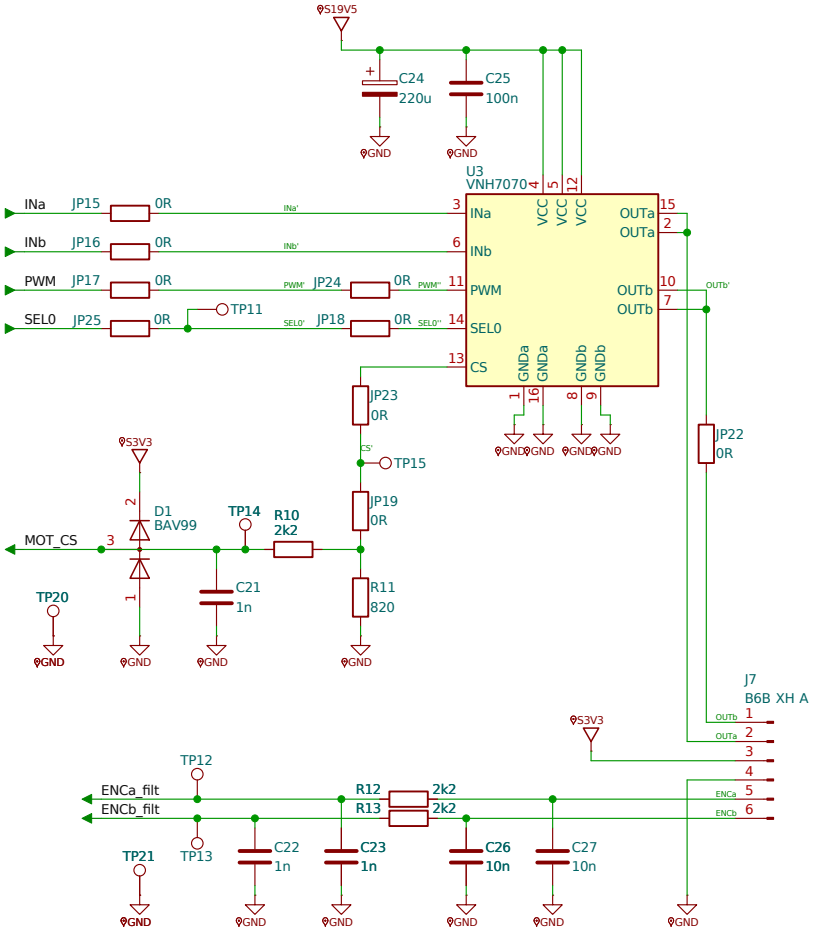
Notes

Sheet: Microcontroller	
File: Microcontroller.kicad_sch	
Rev: 1.1.0	Drwn: FLR
Mat: Concept	Rev'd: FLR
Date: 2026-10-18	Rls'd: FLR
Dwg: NA	Size: A4
Generated with KiCad	Page: 8/11

Servomotor interface

Block interface



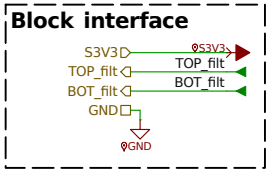


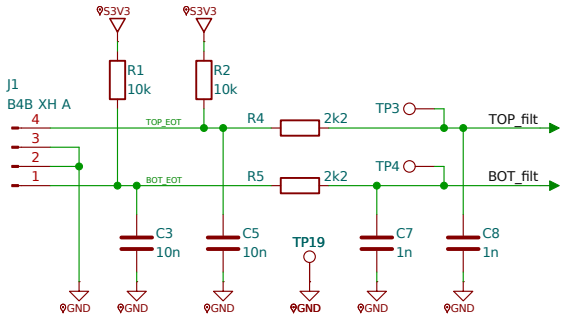
Return to front sheet

ChurrasCod Electronics	**Desk controller**	Sheet: Servomotor interface
	Notes	File: Servomotor_interface.kicad_sch
		Rev: 1.1.0 / Drwn: FLR
		Mat: Concept / Rev'd: FLR
		Date: 2026-10-18 / Rls'd: FLR
		Dwg: NA / Size: A4
		Generated with KiCad / Page: 9/11

Limit switch interface

Block interface





Return to front sheet

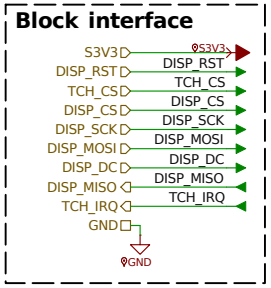
ChurrasCod Electronics

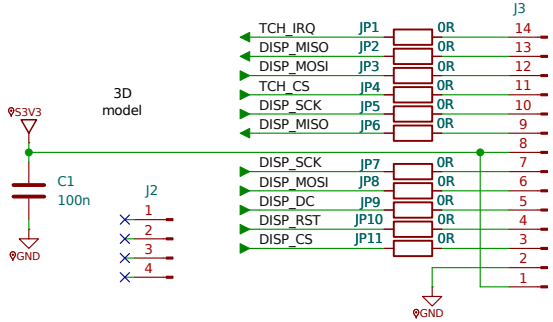
Desk controller

Notes

Sheet: Limit switch interface	
File: Limit_switch_interface.kicad_sch	
Rev: 1.1.0	Drwn: FLR
Mat: Concept	Rev'd: FLR
Date: 2026-10-18	Rls'd: FLR
Dwg: NA	Size: A4
Generated with KiCad	Page: 10/11

Display





TFT1
TFT_2.4in_320x240

The display will not be directly soldered to the PCB. Instead, generic 2.54 mm pitch sockets J2 and J3 will be used as connectors.

Return to front sheet

ChurrasCod Electronics

Desk controller

Notes

Sheet: Display	
File: Display.kicad_sch	
Rev: 1.1.0	Drwn: FLR
Mat: Concept	Rev'd: FLR
Date: 2026-10-18	Rls'd: FLR
Dwg: NA	Size: A4
Generated with KiCad	Page: 11/11